

Mert Kosan

STAFF MACHINE LEARNING SCIENTIST

Austin, TX, USA

✉ mertkosan@gmail.com | 🏠 mertkosan.com | 🔗 mertkosan | 📄 mertkosan | 📖 Papers | 📄 Patents

Professional Experience

Visa Inc.

Austin, TX, USA

Staff Machine Learning Scientist

June 2023 – Present

- Achieved a 3–4% fraud capture rate (FCR) lift over current production models in early benchmarks by designing and developing large-scale self-supervised transformers and Large Transactional Models (LTMs) for multi-task fraud detection, scam detection, risk modeling, and payment intelligence; progressing toward deployment.
- Lifted FCR and value detection rate (VDR) while lowering false positive rate (FPR) to client-required levels by building real-time fraud detection models for Visa Direct Original Credit Transactions (OCT), using large-scale transactional, sequential, and behavioral data, in collaboration with engineering and product teams.
- Drove over \$250M in potential annual fraud capture across multiple merchants by designing the full feature and signal architecture for an account-level insights system — including first- and third-party fraud signals, risk scores, and features across billions of accounts — and partnering with cross-functional stakeholders to bring it into production.
- Enhanced Visa Real-Time Payments Prevent with a GenAI-powered explainability system that generates human-readable rationale for high-risk-score transaction decisions, deployed internally to support model transparency and human-in-the-loop decision-making.
- Mentored interns and machine learning scientists across multiple projects, guiding technical direction and experimental design to translate research ideas into production-ready solutions.

Visa Research

Austin, TX, USA

Ph.D. Research Intern

Summers 2020, 2021, and 2022

- Developed a human-in-the-loop active learning framework that models expert interests and expertise to improve data labeling, anomaly detection, and AI-assisted decision-making.
- Designed a scalable and parallelizable dynamic peer-group analysis framework compatible with existing anomaly detection methods, reducing false positives and enabling earlier detection of anomalous events.
- Developed a graph-based framework for automated fraud profiling and strategy optimization in near-real-time unsupervised fraud detection systems, generating accurate fraud profiles in under one second from limited transaction data.

PhD Research Highlights

University of California, Santa Barbara

Santa Barbara, CA, USA

GNNX-BENCH: Benchmarking Perturbation-Based GNN Explainers

March 2023 – June 2023

- Led a comprehensive evaluation of seven factual and four counterfactual graph neural network explainers across multiple datasets, model architectures, and evaluation criteria.
- Identified key limitations in explanation stability, adversarial robustness, reproducibility, and predictive utility, and released an open-source benchmarking framework to support reproducible research.

University of California, Santa Barbara

Santa Barbara, CA, USA

GCFExplainer: Global Counterfactual Explanations for Graph Neural Networks

October 2021 – August 2022

- Formulated the problem of global counterfactual explanation for graph classification and developed GCFExplainer, a method that generates representative counterfactual summaries using vertex-reinforced random walks over an edit map.
- Achieved state-of-the-art performance across multiple graph classification benchmarks; the work was selected among the Top 10 papers at WSDM 2023 and received the Best Paper Award at MLoG-WSDM 2023.

Education

University of California, Santa Barbara

Santa Barbara, CA, USA

Ph.D. and M.S. in Computer Science

September 2018 – June 2023

- Advisor: Prof. Ambuj K. Singh | GPA: 4.00/4.00.
- Dissertation: *Transparent Representation Learning for Graphs and Human-AI Collaboration*.

Sabanci University

Istanbul, Turkey

B.S. in Computer Science

September 2013 – June 2018

- Ranked first in the program; GPA: 3.99/4.00.

Patents

- **Improving Machine Learning Models** (2024) – Multi-model pipeline that learns entity and user-specific embeddings, extracts features, and generates relevance scores, using a human-in-the-loop feedback signal to continuously retrain the models.
- **Dynamic Peer Group Analysis of Systematic Changes in Large-Scale Data** (2023) – Graph-based method that groups similar entities into peer groups and compares their risk-score statistics to distinguish false anomalies (e.g. seasonal spikes) from true anomalies, reducing false positives and enabling earlier detection.
- **Auto-Profiling Anomalies** (2022) – System that automatically clusters flagged anomaly transactions by weighted feature importance and labels each cluster with a feature profile, enabling real-time classification of new anomalies without manual review.

Selected Publications

- [KDD 2025] *Attribute-Enhanced Similarity Ranking for Sparse Link Prediction*.
- [ACM TIST 2024] *GCFExplainer: Global Counterfactual Explainer for Graph Neural Networks*.
- [KDD 2024] *Global Human-Guided Counterfactual Explanations for Molecular Properties via Reinforcement Learning*.
- [ICLR 2024] *GNNX-BENCH: Unravelling the Utility of Perturbation-Based GNN Explainers through In-Depth Benchmarking*.
- [AAAI 2024] *DGCLUSTER: A Neural Framework for Attributed Graph Clustering via Modularity Maximization*.
- [WSDM 2023] *Global Counterfactual Explainer for Graph Neural Networks*. **Top 10 at WSDM; Best Paper at MLoG-WSDM**.

250+ citations, h-index 7 (Google Scholar).

Academic Service

- **Leadership:** Area Chair, KDD 2025 Dataset and Benchmark Track (nine meta-reviews); Registration Chair, KDD 2023.
- **Reviewer:** More than 35 reviews for ICLR, NeurIPS, KDD, The Web Conference, ICDE, AAMAS, TNNLS, TIST, TKDD, TPAMI, and TAI.

Technical Skills

Machine Learning	Transformers, Graph Neural Networks, Representation Learning, Anomaly Detection, Explainable AI, Generative AI
Fraud and Risk	Fraud and Scam Detection, Real-Time Payments, Behavioral Modeling
Big Data	Spark, PySpark, Hadoop, Hive
Programming	Python, PyTorch, PyTorch Geometric, Scikit-learn, SQL, C++, Java
Tools	Linux, Git, CUDA, NetworkX, Airflow